

Stochastic State-Space SIRS/SEIRS Models

Final Project Report

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1 Biological Problem

SIR and SEIR models are used to model the spread of a pathogen as people move between susceptible, exposed (for SEIR models), infected, and recovered states. SIR and SEIR models help us understand how pathogens will spread and how they can be controlled, making them one of the most relevant mathematical models in biology. We aim to fit our models to a dataset on the COVID-19 pandemic caused by the SARS-CoV-2 virus for each region of Africa, Asia, Europe, North America, South America, Oceania, and three global replicates, and add complexity to our models as we account for more biological realism and compare how well each of our models converge. The virus is spread through close contact between infected and susceptible individuals, as infected individuals breathe out droplets and small particles containing the virus (CDC, 2024). Previous studies have found COVID-19 to have a mean latent period of five days, mean infectious period of seven days, and mean immune duration of 180 days (Johns Hopkins Medicine, 2022). New genomic variants of SARS-CoV-2, especially the Omicron variant, have dynamic effects (usually positive) on the transmission of COVID-19 (CDC, 2024).

Deterministic versions of SIR and SEIR models with fixed rates of infection, incubation, recovery, and waning immunity are very useful for epidemiology but do not account for the inherent randomness or dynamic nature of real world epidemics. To account for this, we aim to add stochasticity, elements that are randomly determined that cannot be predicted, to our models. We also allow the transmission rate to vary based on the external factors of genomic variants and government stringency. New genomic

variants of a virus will affect its transmission rate, as well as changes in government policy such as school closures or restrictions on public gatherings, which should have a negative effect on transmission rates (Roser, 2021). Our later models aim to incorporate greater biological realism by modeling the infectious and latent periods using Erlang distributions, where states are broken into sub-compartments that individuals flow through, which better model the progression of disease compared to the memoryless exponential assumption. Additionally, we aim to incorporate partial immunity into our model, as in the real world recovered individuals may have partial immunity letting them become reinfected or exposed directly from the recovered population (Biswas et al., 2026).

2 Assumptions

2.1 M1 SIRS

We assume that each population is closed, meaning that there are no births, deaths, or movement between regions, and the total population is split into a susceptible, infected, and recovered population. New infections have a chance to occur when susceptible individuals come in contact with infected individuals. The force of infection, or rate at which the susceptible population becomes infected, is affected by genomic volatility signal of the previous rate and the current government stringency index. Infected individuals have a fixed probability of recovering. Recovered individuals have a fixed probability of waning immunity and becoming susceptible again. The vaccination rate is proportional to the entire population N but only susceptible individuals get vaccinated, after which they enter the recovered population.

2.2 M2 SEIRS

We instead assume that the population is split into susceptible, exposed, infected, and recovered populations. After contact with infected individuals, susceptible individuals have a chance to become exposed. Exposed individuals have a fixed probability of becoming infected.

2.3 M3 SIRS / M4 SEIRS

Instead of using the genomic volatility of the previous week, we assume that genomic volatility accumulates over time and decays at a fixed rate.

2.4 M5 SIRS

We instead assume that the time spent in the infected population follows an Erlang distribution with $k=4$. This splits the infected population into four sub-compartments that each individual must move through before recovering.

2.5 M6 SEIRS

We instead assume that the time spent in the exposed population follows an Erlang distribution with $k=4$. This splits the exposed population into four sub-compartments that each individual must move through before becoming infected.

2.6 M7 SIRS / M8 SEIRS

We assume that some fraction of the recovered population will have partial immunity and return to infected for M7 and exposed for M8 instead of susceptible.

3 Parameter Definitions

3.1 Data Inputs

Symbol	Definition
y_t	Observed weekly COVID-19 cases
$\nu_{lag,t}$	Genomic volatility signal, pre-lagged one week prior to model fitting (M1, M2)
$\nu_{signal,t}$	Raw standardized genomic volatility signal (M3-M8)
Z_t	Standardized Government Stringency Index
$\nu_{vax,t}$	Weekly incident vaccination proportion of population
N	Total regional population (fixed)
ρ	Region-specific case reporting rate (fixed)
σ	Incubation rate = $1/5 \text{ wk}^{-1}$; mean latent period 5 days (SEIR models; fixed)
γ	Recovery rate = $1/7 \text{ wk}^{-1}$; mean infectious period 7 days (fixed)
ω	Waning immunity rate = $1/180 \text{ wk}^{-1}$; mean immune duration 180 days (fixed)
k	Erlang shape parameter = 4; controls variance of dwell-time distribution (M5–M8; fixed)
η	Partial immunity fraction = 0.5; proportion of waning R bypassing full susceptibility (M7, M8; fixed)

3.2 Estimated Parameters

Symbol	Prior	Models	Definition
β_0	LogNormal(0, 0.5)	All	Baseline transmission rate
α_ν	Normal(0, 0.3)	All	Sensitivity of transmission to genomic volatility; positive values indicate variant pressure amplifies transmission
α_{str}	Normal(0, 0.3)	Full only	Sensitivity of transmission to government stringency; expected to be negative
s_0	Beta(3, 3)	All	Initial susceptible fraction; $S_1 = N \cdot s_0$
i_0	Beta(1, 80)	All	Initial infectious fraction; $I_1 = N \cdot i_0$
e_0	Beta(1, 80)	SEIR only	Initial exposed fraction; $E_1 = N \cdot e_0$
δ	Beta(2, 5)	M3–M8	Genomic memory decay rate; small δ reflects slow immune adaptation, large δ reflects dominance of recent variant pressure
ϕ^{-1}	Gamma(2, 0.1)	All	Inverse overdispersion; accounts for the overdispersed clumping typical of COVID-19 case data

3.3 Derived Quantities

Symbol	Definition
β_t	Time-varying transmission rate = $\beta_0 \exp(\alpha_\nu \Theta_t + \alpha_{str} Z_t)$
λ_t	Force of infection = $\beta_t \cdot I_t / N$
A_t	Genomic volatility memory state = $(1 - \delta)A_{t-1} + \nu_{signal,t}$; accumulates variant pressure over time (M3–M8)
Θ_t	Active forcing signal: pre-lagged $\nu_{lag,t}$ for M1/M2; memory state A_t for M3–M8
Δ_{vax}	Weekly flow of vaccinated individuals out of S : $\min(S_t, \nu_{vax,t} \cdot N)$
S_t^*	Post-vaccination susceptible pool = $S_t - \Delta_{vax}$
γ_k	Per-stage Erlang recovery rate = $k\gamma$
σ_k	Per-stage Erlang incubation rate = $k\sigma$
μ_t	Expected observed cases; new infections scaled by reporting rate ρ (model-dependent)

4 Model Ladder

ID	Structure	E comp.	Erlang ($k = 4$)	Forcing	Waning
M1	SIRS	–	–	Lag ν_{lag}	Full $R \rightarrow S$
M2	SEIRS	Yes	–	Lag ν_{lag}	Full $R \rightarrow S$
M3	SIRS	–	–	Memory A_t	Full $R \rightarrow S$
M4	SEIRS	Yes	–	Memory A_t	Full $R \rightarrow S$
M5	SIRS	–	I only	Memory A_t	Full $R \rightarrow S$
M6	SEIRS	Yes	E & I	Memory A_t	Full $R \rightarrow S$
M7	SIRS	–	I only	Memory A_t	Partial $\eta = 0.5 \rightarrow I_1$
M8	SEIRS	Yes	E & I	Memory A_t	Partial $\eta = 0.5 \rightarrow E_1$

5 Shared Structural Elements

5.1 Genomic Forcing

Tier 1 (M1, M2) uses a pre-lagged signal; Tier 2–4 (M3–M8) replaces it with a learnable memory state A_t :

$$A_t = (1 - \delta) A_{t-1} + \nu_{signal,t}, \quad A_1 = \nu_{signal,1} \quad (1)$$

$$\lambda_t = \frac{\beta_0 \exp(\min[\alpha_\nu \Theta_t + \alpha_{str} Z_t, 10]) \cdot I_t}{N} \quad (2)$$

where $\Theta_t = \nu_{lag,t}$ (M1, M2) or $\Theta_t = A_t$ (M3–M8), Z_t is the standardised stringency index, and $I_t = \sum_j I_{t,j}$ for Erlang models.

5.2 Vaccination Deduction

$$\Delta_{vax} = \min(S_t, \nu_{vax,t} \cdot N), \quad S_t^* = S_t - \Delta_{vax} \quad (3)$$

5.3 Discrete Transition Probabilities

Continuous-time rates are converted to weekly probabilities via the exact survival probability $1 - e^{-r}$, ensuring flows remain in $[0, 1]$:

$$p_\lambda = 1 - e^{-\lambda_t} \quad (\text{new infections}) \quad (4)$$

$$p_\sigma = 1 - e^{-\sigma} \quad (\text{incubation, SEIR exponential}) \quad (5)$$

$$p_\gamma = 1 - e^{-\gamma} \quad (\text{recovery, exponential}) \quad (6)$$

$$p_{\sigma_k} = 1 - e^{-k\sigma} \quad (\text{per-stage incubation, Erlang}) \quad (7)$$

$$p_{\gamma_k} = 1 - e^{-k\gamma} \quad (\text{per-stage recovery, Erlang}) \quad (8)$$

$$p_\omega = 1 - e^{-\omega} \quad (\text{waning}) \quad (9)$$

5.4 Observation Model

$$y_t \sim \text{NegBin}_2(\mu_t, \phi^{-1}), \quad \mu_t = \begin{cases} \Delta_{SI} \cdot \rho & \text{SIRS backbone (M1, M3, M5)} \\ (\Delta_{SI} + \Delta_{RI}) \cdot \rho & \text{M7 (partial immunity)} \\ \Delta_{EI} \cdot \rho & \text{SEIRS exponential (M2, M4)} \\ \Delta_{E_4} \cdot \rho & \text{SEIRS Erlang (M6, M8)} \end{cases} \quad (10)$$

6 ODE Skeletons and Discrete State Updates

6.1 M1 / M2 — Exponential SIR(S) and SEIR(S), Lag Forcing

ODE skeleton (M1 SIRS):

$$\dot{S} = \omega R - \lambda_t S - \nu_{vax} N$$

$$\dot{I} = \lambda_t S - \gamma I$$

$$\dot{R} = \gamma I - \omega R + \nu_{vax} N$$

ODE skeleton (M2 SEIRS, adds E):

$$\begin{aligned}\dot{S} &= \omega R - \lambda_t S - \nu_{vax} N \\ \dot{E} &= \lambda_t S - \sigma E \\ \dot{I} &= \sigma E - \gamma I \\ \dot{R} &= \gamma I - \omega R + \nu_{vax} N\end{aligned}$$

Discrete updates (M1):

$$\begin{aligned}\Delta_{SI} &= S_t^*(1 - e^{-\lambda_t}), \quad \Delta_{IR} = I_t(1 - e^{-\gamma}), \quad \Delta_{RS} = R_t(1 - e^{-\omega}) \\ S_{t+1} &= S_t^* - \Delta_{SI} + \Delta_{RS} \\ I_{t+1} &= I_t + \Delta_{SI} - \Delta_{IR} \\ R_{t+1} &= R_t + \Delta_{IR} - \Delta_{RS} + \Delta_{vax}\end{aligned}$$

Discrete updates (M2, adds E layer):

$$\begin{aligned}\Delta_{SE} &= S_t^*(1 - e^{-\lambda_t}), \quad \Delta_{EI} = E_t(1 - e^{-\sigma}), \quad \Delta_{IR} = I_t(1 - e^{-\gamma}), \quad \Delta_{RS} = R_t(1 - e^{-\omega}) \\ S_{t+1} &= S_t^* - \Delta_{SE} + \Delta_{RS} \\ E_{t+1} &= E_t + \Delta_{SE} - \Delta_{EI} \\ I_{t+1} &= I_t + \Delta_{EI} - \Delta_{IR} \\ R_{t+1} &= R_t + \Delta_{IR} - \Delta_{RS} + \Delta_{vax}\end{aligned}$$

6.2 M3 / M4 — Memory Kernel, Exponential Dwell Times

M3 and M4 are structurally identical to M1 and M2 respectively, with $\nu_{lag,t}$ replaced by the memory state A_t in the forcing exponent. All discrete updates are identical. ODE skeletons add:

$$\dot{A} = \nu_{signal} - \delta A$$

6.3 M5 / M6 — Erlang-Distributed Dwell Times ($k = 4$)

ODE skeleton (M5, Erlang I):

$$\begin{aligned}\dot{A} &= \nu_{signal} - \delta A \\ \dot{S} &= \omega R - \lambda_t S - \nu_{vax} N \\ \dot{I}_1 &= \lambda_t S - k\gamma I_1 \\ \dot{I}_j &= k\gamma I_{j-1} - k\gamma I_j, \quad j \in \{2, 3, 4\} \\ \dot{R} &= k\gamma I_4 - \omega R + \nu_{vax} N\end{aligned}$$

ODE skeleton (M6, Erlang E and I):

$$\begin{aligned}
\dot{A} &= \nu_{signal} - \delta A \\
\dot{S} &= \omega R - \lambda_t S - \nu_{vax} N \\
\dot{E}_1 &= \lambda_t S - k\sigma E_1 \\
\dot{E}_j &= k\sigma E_{j-1} - k\sigma E_j, \quad j \in \{2, 3, 4\} \\
\dot{I}_1 &= k\sigma E_4 - k\gamma I_1 \\
\dot{I}_j &= k\gamma I_{j-1} - k\gamma I_j, \quad j \in \{2, 3, 4\} \\
\dot{R} &= k\gamma I_4 - \omega R + \nu_{vax} N
\end{aligned}$$

Discrete updates (M5):

$$\begin{aligned}
\Delta_{SI} &= S_t^*(1 - e^{-\lambda_t}), \quad \Delta_{I_j} = I_{t,j}(1 - e^{-k\gamma}), \quad \Delta_{RS} = R_t(1 - e^{-\omega}) \\
S_{t+1} &= S_t^* - \Delta_{SI} + \Delta_{RS} \\
I_{t+1,1} &= I_{t,1} + \Delta_{SI} - \Delta_{I_1} \\
I_{t+1,j} &= I_{t,j} + \Delta_{I_{j-1}} - \Delta_{I_j}, \quad j \in \{2, 3, 4\} \\
R_{t+1} &= R_t + \Delta_{I_4} - \Delta_{RS} + \Delta_{vax}
\end{aligned}$$

Discrete updates (M6, adds Erlang E):

$$\begin{aligned}
\Delta_{SE} &= S_t^*(1 - e^{-\lambda_t}), \quad \Delta_{E_j} = E_{t,j}(1 - e^{-k\sigma}), \quad \Delta_{I_j} = I_{t,j}(1 - e^{-k\gamma}), \quad \Delta_{RS} = R_t(1 - e^{-\omega}) \\
S_{t+1} &= S_t^* - \Delta_{SE} + \Delta_{RS} \\
E_{t+1,1} &= E_{t,1} + \Delta_{SE} - \Delta_{E_1} \\
E_{t+1,j} &= E_{t,j} + \Delta_{E_{j-1}} - \Delta_{E_j}, \quad j \in \{2, 3, 4\} \\
I_{t+1,1} &= I_{t,1} + \Delta_{E_4} - \Delta_{I_1} \\
I_{t+1,j} &= I_{t,j} + \Delta_{I_{j-1}} - \Delta_{I_j}, \quad j \in \{2, 3, 4\} \\
R_{t+1} &= R_t + \Delta_{I_4} - \Delta_{RS} + \Delta_{vax}
\end{aligned}$$

6.4 M7 / M8 — Partial Immunity ($\eta = 0.5$)

Waning is split: fraction η bypasses full susceptibility.

ODE skeleton (M7, partial $\rightarrow I_1$):

$$\begin{aligned}
\dot{A} &= \nu_{signal} - \delta A \\
\dot{S} &= (1 - \eta)\omega R - \lambda_t S - \nu_{vax} N \\
\dot{I}_1 &= \lambda_t S + \eta\omega R - k\gamma I_1 \\
\dot{I}_j &= k\gamma I_{j-1} - k\gamma I_j, \quad j \in \{2, 3, 4\} \\
\dot{R} &= k\gamma I_4 - \omega R + \nu_{vax} N
\end{aligned}$$

ODE skeleton (M8, partial $\rightarrow E_1$):

$$\begin{aligned}
\dot{A} &= \nu_{signal} - \delta A \\
\dot{S} &= (1 - \eta)\omega R - \lambda_t S - \nu_{vax} N \\
\dot{E}_1 &= \lambda_t S + \eta\omega R - k\sigma E_1 \\
\dot{E}_j &= k\sigma E_{j-1} - k\sigma E_j, \quad j \in \{2, 3, 4\} \\
\dot{I}_1 &= k\sigma E_4 - k\gamma I_1 \\
\dot{I}_j &= k\gamma I_{j-1} - k\gamma I_j, \quad j \in \{2, 3, 4\} \\
\dot{R} &= k\gamma I_4 - \omega R + \nu_{vax} N
\end{aligned}$$

Discrete updates (M7):

$$\begin{aligned}
\Delta_{RS} &= R_t(1 - e^{-\omega})(1 - \eta) \rightarrow S, & \Delta_{RI} &= R_t(1 - e^{-\omega})\eta \rightarrow I_{t,1} \\
S_{t+1} &= S_t^* - \Delta_{SI} + \Delta_{RS} \\
I_{t+1,1} &= I_{t,1} + \Delta_{SI} + \Delta_{RI} - \Delta_{I_1} \\
I_{t+1,j} &= I_{t,j} + \Delta_{I_{j-1}} - \Delta_{I_j}, \quad j \in \{2, 3, 4\} \\
R_{t+1} &= R_t + \Delta_{I_4} - \Delta_{RS} - \Delta_{RI} + \Delta_{vax}
\end{aligned}$$

Discrete updates (M8):

$$\begin{aligned}
\Delta_{RS} &= R_t(1 - e^{-\omega})(1 - \eta) \rightarrow S, & \Delta_{RE} &= R_t(1 - e^{-\omega})\eta \rightarrow E_{t,1} \\
S_{t+1} &= S_t^* - \Delta_{SE} + \Delta_{RS} \\
E_{t+1,1} &= E_{t,1} + \Delta_{SE} + \Delta_{RE} - \Delta_{E_1} \\
E_{t+1,j} &= E_{t,j} + \Delta_{E_{j-1}} - \Delta_{E_j}, \quad j \in \{2, 3, 4\} \\
I_{t+1,1} &= I_{t,1} + \Delta_{E_4} - \Delta_{I_1} \\
I_{t+1,j} &= I_{t,j} + \Delta_{I_{j-1}} - \Delta_{I_j}, \quad j \in \{2, 3, 4\} \\
R_{t+1} &= R_t + \Delta_{I_4} - \Delta_{RS} - \Delta_{RE} + \Delta_{vax}
\end{aligned}$$

7 Sensitivity Variants

Ablation analysis was performed on models by removing α_{str} and vaccination to see what effect this has on α_ν .

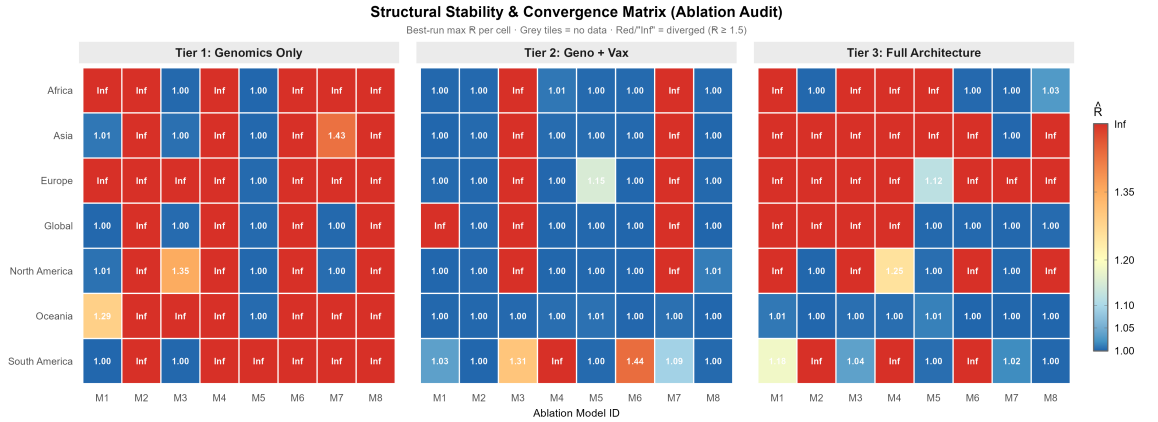
$$\lambda_t^{naked} = \frac{\beta_0 \exp(\alpha_\nu \Theta_t) \cdot I_t}{N} \quad \Theta_t = \begin{cases} \nu_{lag,t} & \text{M1 and M2 Genomics and Genomics + vaccination} \\ A_t & \text{M3 - M8 Genomics and Genomics+vaccination} \end{cases} \quad (11)$$

Purpose: This was done to isolate the genomic forcing from the policy decisions and vaccination in order to see what effect that has on the mean α_ν values.

8 Simulation

Bayesian posterior estimation and trajectories were conducted using cmdstan software package on r which uses the No-U-turn sampler. This is a version of the Hamiltonian Monte Carlo method which allows quantification of uncertainty. Models that were accepted for final analysis were ones that showed a high level of convergence ($\hat{R} < 1.1$).

9 Results & Analysis



These Heatmaps show which of the individual model types and architectures converged for each of the complexity levels given a total of 485 runs. It is possible that with more runs we could be able to get more of these to converge. The trends are that more convergence happened on tier 2 overall, for

tier 3 the more complex Erlang based memory models tended to converge. Odd Models appeared to do better on tier 1 and 2 more generally.

Table 1. Tier 1: Genomics Only — Posterior Parameter Estimates								
Posterior median [90% CI] from best-converged run per region $\times$ parameter ¹								
	α_v — Genomic effect		β_0 — Baseline transmission		δ — Memory decay rate		1/OD — Overdispersion	
	Median [90% CI]	$\hat{R}^2$	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$
Africa	0.197 [0.114, 0.295]	1.000	0.111 [0.102, 0.120]	1.000	0.447 [0.253, 0.662]	1.000	1.904 [1.580, 2.275]	1.000
Asia	0.653 [0.506, 0.810]	1.001	1.494 [1.070, 2.131]	1.001	0.252 [0.194, 0.338]	1.000	0.148 [0.134, 0.164]	1.000
Europe	0.040 [0.028, 0.070]	1.002	0.097 [0.084, 0.109]	1.000	0.041 [0.022, 0.103]	1.002	0.611 [0.543, 0.684]	1.000
Global	0.018 [0.016, 0.021]	1.000	0.253 [0.238, 0.275]	1.000	0.017 [0.014, 0.022]	1.001	1.175 [1.046, 1.322]	1.001
North America	0.068 [0.048, 0.083]	1.000	0.362 [0.265, 0.470]	1.000	0.020 [0.014, 0.028]	1.000	0.292 [0.261, 0.324]	1.001
Oceania	-0.115 [-0.169, -0.086]	1.000	0.126 [0.100, 0.150]	1.002	0.030 [0.021, 0.041]	1.002	1.120 [0.905, 1.355]	1.002
South America	0.206 [0.127, 0.304]	1.001	4.039 [3.028, 5.328]	1.001	0.327 [0.211, 0.495]	1.000	0.575 [0.488, 0.673]	1.000

¹ Oceania $\alpha_v = -0.115$ [-0.169, -0.087]: only negative genomic effect at Tier 1. South America $\beta_0 = 4.04$ — inflated without vaccination data.

² $\hat{R} > 1.05$ (yellow) = marginal convergence.

Best-run selection: minimum max- $\hat{R}$ per run.

For the Genomics only runs all regions besides Oceania show strong genomic coupling as α_v is positive throughout the regions. South Americas genomic impact at this tier of the ablation analysis appears to be overinflated.

Table 2. Tier 2: Genomics + Vaccination — Posterior Parameter Estimates								
Posterior median [90% CI] from best-converged run per region $\times$ parameter ¹								
	α_v — Genomic effect		β_0 — Baseline transmission		δ — Memory decay rate		1/OD — Overdispersion	
	Median [90% CI]	$\hat{R}^2$	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$
Africa	0.587 [0.499, 0.668]	1.001	0.493 [0.380, 0.634]	1.000	0.382 [0.123, 0.675]	1.000	2.379 [1.967, 2.855]	1.001
Asia	-0.314 [-0.569, -0.111]	1.000	0.173 [0.139, 0.220]	1.000	0.168 [0.051, 0.428]	1.000	0.090 [0.082, 0.100]	1.000
Europe	0.093 [-0.131, 0.379]	1.000	0.233 [0.157, 0.396]	1.000	0.336 [0.106, 0.618]	1.000	0.419 [0.374, 0.467]	1.000
Global	0.201 [0.055, 0.376]	1.000	0.297 [0.181, 0.671]	1.000	0.108 [0.042, 0.437]	1.000	0.324 [0.291, 0.359]	1.000
North America	0.436 [0.407, 0.467]	1.000	0.210 [0.202, 0.221]	1.000	0.192 [0.080, 0.462]	1.000	0.560 [0.501, 0.626]	1.000
Oceania	-0.030 [-0.235, 0.125]	1.000	0.779 [0.693, 0.835]	1.001	0.024 [0.009, 0.043]	1.000	1.076 [0.878, 1.302]	1.000
South America	0.514 [0.235, 0.823]	1.000	0.522 [0.313, 0.860]	1.000	0.303 [0.185, 0.503]	1.001	0.316 [0.271, 0.366]	1.000

¹ Asia α_v reverses sign vs. Tier 1 (+0.654 $\rightarrow$ -0.314): vaccination data substantially revises inferred genomic effect direction.

² $\hat{R} > 1.05$ (yellow) = marginal convergence.

Best-run selection: minimum max- $\hat{R}$ per run.

When adding in vaccination to the ablation analysis the genomic cou-

pling effect decreases across most regions, even to the point of flipping to negative in Asia. Overall the genomic coupling effect is positive across most regions. Oceania’s genomic coupling effect became more positive at this tier level.

Table 3. Tier 3: Full Architecture — Posterior Parameter Estimates										
Adds structural covariate α_{str} (Tier 3 only). Posterior median [90% CI], best-converged run.										
	α_v		β_0		δ		$1/OD^{\dagger}$		α_{str} (Tier 3 only)	
	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$	Median [90% CI]	$\hat{R}$
Africa	-0.483 [-0.752, -0.261]	1.000	1.223 [0.570, 2.613]	1.001	0.326 [0.217, 0.510]	1.001	1.982 [1.655, 2.382]	1.001	-0.033 [-0.048, -0.019]	1.001
Asia	-0.258 [-0.286, -0.225]	1.000	4.865 [2.697, 8.705]	1.001	0.023 [0.019, 0.028]	1.000	0.010 [0.010, 0.010]	1.000	-0.062 [-0.072, -0.053]	1.000
Europe	0.260 [0.108, 0.742]	1.044	0.773 [0.320, 1.843]	1.005	0.023 [0.009, 0.269]	1.045	0.419 [0.374, 0.469]	1.002	-0.066 [-0.138, 0.199]	1.021
Global	0.227 [0.049, 0.460]	1.000	0.312 [0.164, 0.610]	1.001	0.231 [0.075, 0.521]	1.001	0.324 [0.290, 0.359]	1.000	-0.005 [-0.016, 0.006]	1.000
North America	0.292 [0.103, 0.623]	1.000	0.497 [0.247, 1.089]	1.000	0.194 [0.081, 0.458]	1.001	0.175 [0.157, 0.193]	1.000	0.001 [-0.012, 0.016]	1.000
Oceania	-0.004 [-0.498, 0.492]	1.000	0.919 [0.397, 2.061]	1.000	0.267 [0.062, 0.565]	1.000	1.019 [0.831, 1.232]	1.000	-0.385 [-0.690, -0.232]	1.000
South America	0.351 [0.100, 0.609]	1.000	1.103 [0.488, 2.421]	1.001	0.274 [0.136, 0.533]	1.000	0.010 [0.010, 0.010]	1.001	-0.018 [-0.032, -0.004]	1.000
[†] Pink $1/OD$ = lower-bound hit (0.010); overdispersion unidentifiable.										
Best-run selection: minimum max- $\hat{R}$ per run.										

Africa and Asia are both negative at this tier level. Oceania is near 0 at this level and credible interval passing 0 in this tier. Over-dispersion is unidentifiable in some of these regions, South America and Asia in particular.

Table 4. Volterra Memory Kernel Half-Lives						
t½ = ln(2)/δ × 7 days (weekly units). Best-converged run per region × tier.						
	δ (decay rate, per week)		t½ (days) ¹		t½ (wks)	R̂
	Median	90% CI	Median	90% CI		
Tier 1						
Africa	0.4468	[0.2528, 0.6623]	10.9	[7.3, 19.2]	1.55	1.000
Asia	0.2519	[0.1940, 0.3376]	19.3	[14.4, 25.0]	2.75	1.000
Europe	0.0409	[0.0223, 0.1028]	118.7	[47.2, 217.2]	16.96	1.002
Global	0.0170	[0.0141, 0.0222]	284.7	[218.7, 344.4]	40.68	1.001
North America	0.0200	[0.0143, 0.0284]	243.0	[170.9, 340.3]	34.71	1.000
Oceania	0.0298	[0.0214, 0.0406]	162.8	[119.6, 227.0]	23.25	1.002
South America	0.3266	[0.2107, 0.4948]	14.9	[9.8, 23.0]	2.12	1.000
Tier 2						
Africa	0.3822	[0.1231, 0.6755]	12.7	[7.2, 39.4]	1.81	1.000
Asia	0.1678	[0.0510, 0.4281]	28.9	[11.3, 95.2]	4.13	1.000
Europe	0.3363	[0.1056, 0.6177]	14.4	[7.9, 45.9]	2.06	1.000
Global	0.1079	[0.0419, 0.4369]	45.0	[11.1, 115.9]	6.42	1.000
North America	0.1923	[0.0800, 0.4616]	25.2	[10.5, 60.7]	3.60	1.000
Oceania	0.0240	[0.0093, 0.0432]	201.9	[112.4, 519.2]	28.85	1.000
South America	0.3026	[0.1847, 0.5032]	16.0	[9.6, 26.3]	2.29	1.001
Tier 3						
Africa	0.3262	[0.2170, 0.5097]	14.9	[9.5, 22.4]	2.12	1.001
Asia	0.0228	[0.0189, 0.0277]	212.4	[175.4, 256.4]	30.34	1.000
Europe	0.0232	[0.0087, 0.2720]	209.1	[17.8, 555.9] ²	29.87	1.015
Global	0.2313	[0.0755, 0.5213]	21.0	[9.3, 64.3]	3.00	1.001
North America	0.1938	[0.0814, 0.4584]	25.0	[10.6, 59.6]	3.58	1.001
Oceania	0.2674	[0.0625, 0.5647]	18.1	[8.6, 77.6]	2.59	1.000
South America	0.2741	[0.1358, 0.5328]	17.7	[9.1, 35.7]	2.53	1.000

¹ ● Orange = short memory (t½ < 21 d). ● Blue = long memory (t½ > 60 d).

² Europe Tier 3 CI = [17.8, 555.9 d]; δ near-flat, R̂ = 1.015; treat with caution.

dA/dt = nu_t - δA(t), giving t½ = ln(2)/δ.

¹ ● Orange = short memory ($t_{1/2} < 21$ d). ● Blue = long memory ($t_{1/2} > 60$ d).

² Europe Tier 3 CI = [17.8, 555.9 d]: δ near-flat, $\hat{R} = 1.015$; treat with caution.

$dA/dt = nu_t - \delta A(t)$, giving $t_{1/2} = \ln(2)/\delta$.

Table 4 shows the length of the genomic memory coefficient (δ) for each of the regions they vary widely by tier level and continent. Europe on tier 3 has a very wide credible interval (17.8-555.9 days) so this should be used with caution and may be caused by other confounding variables impacting this result. A significant amount of those with tighter credible intervals are between 3 - 6 weeks overall. This suggests that there is large scale heterogeneity among the regions on how long genomic memory persists.

Table 5. Cross-Tier Summary: α_v and Memory Half-Life							
Posterior medians. Sign reversals and $t^{1/2}$ shifts diagnose model-complexity sensitivity.							
	α_v (genomic effect) [†]			$t^{1/2}$ (memory half-life)			Key finding
	Tier 1	Tier 2	Tier 3	Tier 1	Tier 2	Tier 3	
Africa	+0.197	+0.587	-0.483	10.9 d	12.7 d	14.9 d	α_v sign flip T2→T3; short memory stable
Asia	+0.654	-0.314	-0.258	19.3 d	28.9 d	212.4 d	α_v flips T1→T2; δ collapses at Tier 3
Europe	+0.040	+0.093	+0.260	118.7 d	14.4 d	212.8 d	Large $t^{1/2}$ swing; wide CI at Tier 3
Global	+0.018	+0.201	+0.227	284.7 d	45.0 d	21.0 d	Monotone $t^{1/2}$ compression T1→T3
North America	+0.068	+0.436	+0.292	243.0 d	25.2 d	25.0 d	Tier 1 inflated; T2/T3 stable
Oceania	-0.115	-0.030	-0.004	162.8 d	201.9 d	18.1 d	Negative α_v all tiers; $t^{1/2}$ collapses at T3
South America	+0.206	+0.514	+0.351	14.9 d	16.0 d	17.7 d	Most stable — consistent short memory

[†] Red α_v = negative genomic effect on transmission.

Table 5 shows comparison across tier in each region for both α_v and memory half life. In some regions adding α_{str} lowered the genomic memory coefficient and therefore lowered $t^{1/2}$ but in other regions the effect is reversed and memory increases. This does not seem to coincide strictly with the regions that have a reversal of α_v . Africa and South America have stable short term memories across tiers.

10 Discussion

10.1 Convergence and Model Complexity

Tier 1 models showed better convergence in the sir models. This may be because when the other variables are removed in ablation analysis the genomic forcing has to account for the decreasing portion of the curve as well. This is reflected across the 485 runs in the poor identifiability of SEIR models at this tier when only a single forcing signal is present. In more complex models such as in tier 3 the higher level models tended to converge better which is consistent with the Erlang and memory structure providing a smoother parameter space for the sampler.

10.2 Genomic Coupling Effect α_v

The genomic coupling effect was largely positive for most regions but not all across every tier. This may be from strict border controls such as in Oceania or high coverage vaccination campaigns such as in Asia during the large pandemic waves. (Roser, 2021; Mathieu, 2020). Generally tier 1 models

produced lower values for α_ν , suggesting that when not accounting for co-variates you find that the genomic coupling has to account for both the rise and the fall of the waves when policy decisions and vaccination campaigns account for both.

10.3 Memory Half Life

The memory decay parameter δ varied a lot across regions which indicates there isn't universal memory decay parameter for each region. Global and North America replicates memory parameters shrank down when accounting for vaccination to 22-25 days. When adding further complexity at tier 3, δ destabilizes which suggests an identifiability problem with the model after the introduction of α_{str} when trying to estimate both simultaneously.

10.4 Limitations and Future Study

Several limitations constrain our conclusions. Discrete weekly time steps can cause truncation error especially for rapidly changing compartments near peak pandemic waves. The fixed values we used for ω , σ , and γ were taken from early pandemic estimates and in reality they change throughout the different variant periods and therefore may not accurately reflect every time period in the duration of the pandemic (John Hopkins Medicine, 2022). Future analysis should be done to further examine the relationship between genomic change and the memory kernel to better understand that. Another avenue that might be addressed is allowing the flow of people between regions based on the flow of airline passengers with open source flight data to better model the spread of new variants between continents. Another future extension may be allowing η the partial immunity parameter to be estimated rather than fixed like it was for our models.

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